

AMENDMENT NO. 1 JANUARY 2026
TO
IS 3315 : 2024 DIRECT EVAPORATIVE AIR COOLER — SPECIFICATION
(Fourth Revision)

[Foreword, para 3, e)] — Insert the following at the end:

‘f) Compliance with minimum energy efficiency ratio (EER) has been added.’

[Foreword, para 5] — Substitute the following for the existing:

‘For any deviation between the limit specified in this standard and the national regulations, the limits specified in the national regulations shall come primarily prevail. For example, EER is subject to regular revision by appropriate national authority.’

(Page 1, clause 1, notes) — Insert the following at the end:

‘3 The requirements for d.c. motor air cooler and solar powered air coolers are under consideration. Presently, these are out of scope of the standard.’

(Page 1, clause 3.6) — Insert the following at the end:

‘**3.7 Energy Efficiency Ratio (EER)** — Ratio of the sensible cooling capacity to the effective power input to the device at any given set of rating conditions.

NOTE — This definition is applicable to evaporative cooling system only.’

(Page 2, clause 5.4.3) — Insert the following new paragraph at the end:

‘Alternatively, the motor used for driving the fan/blower shall either be capacitor type induction motors or brushless d.c. (BLDC) motor. The capacitor type induction motor shall comply with the requirements given in IS 996 as applicable for fan duty motors.

The measured power input shall not exceed beyond 10 percent of the declared value, and the speed (RPM) shall be within ± 10 percent of the declared value.

In addition, the BLDC motor shall comply with the requirements (respective clauses of IS 996) as specified in **5.9** of IS 1391 (Part 2) except the full load test.’

(Page 2, clause 5.7) — Substitute the following for the existing:

5.7 Switches

The manual operated switches of air coolers shall conform to IS 3854. The manual operated switches which are not covered under IS 3854, shall comply with the following requirements:

- a) Protection against electric shock as per **10** of IS 3854;
- b) Insulation resistance and electric strength as per **16** of IS 3854;
- c) Mechanical strength as per **20** of IS 3854; and
- d) Resistance to heat as per **21** of IS 3854.

NOTE — The typical manual operated switches and their patterns which are not covered under IS 3854 is shown in Fig. 1.

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(Page 2, clause 5.7) — Insert the following at the end:

‘5.8 Electronic Controller

The electronic controller used for safety or operation of air cooler, shall be tested as per 10, 12, 13, 22 and Annex A of IS/IEC 60730-1.’

(Page 3, clause 7.6) — Substitute the following for the existing:

‘7.6 The static pressure difference between the air at the outlet of the air cooler (in the mixing chamber) and the ambient air at the inlet of the air cooler shall be adjusted to zero with the help of exhaust fan and damper.’

[(Page 4, clause 8.2, Table 2, col (3)] — Insert ‘dB’.

[(Page 4, clause 8.2, Table 2, col (4)] — Insert ‘dB’.

(Page 5, clause 8.6.2) — Substitute the following for the existing:

‘8.6.2 Test Condition — Refer to Step 1 of 8.6.3.’

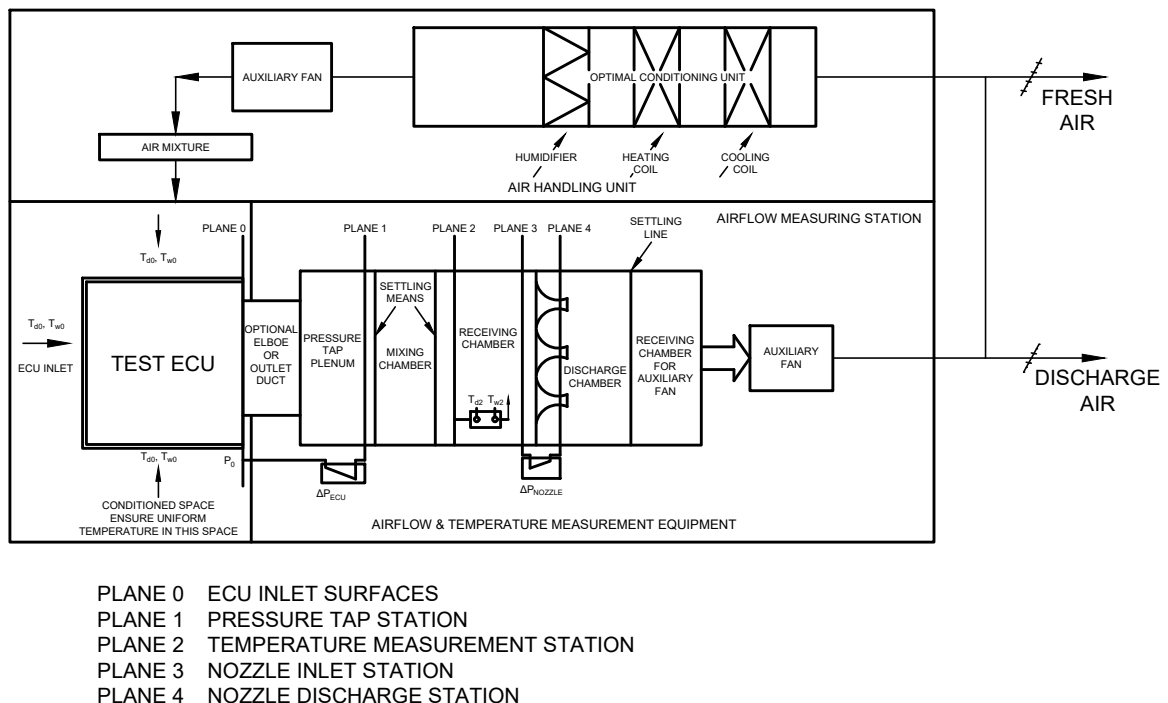
(Page 6, clause 8.6.3, Step 1) — Substitute ‘cooling efficiency’ for ‘evaporation efficiency’.

(Page 6, clause 8.6.3, Step 1, Sl No.5) — Substitute the following for the existing:

‘Rated conditions for inlet air: 40 °C DBT, 24 °C WBT. (see Annex C)’

(Page 7, clause 10.1) — Substitute ‘Fig. 2, Fig. 3 and Fig. 4’ for ‘Fig. 1, Fig. 2 and Fig. 3’, respectively.

(Page 7, Fig. 1) — Substitute the following for the existing:

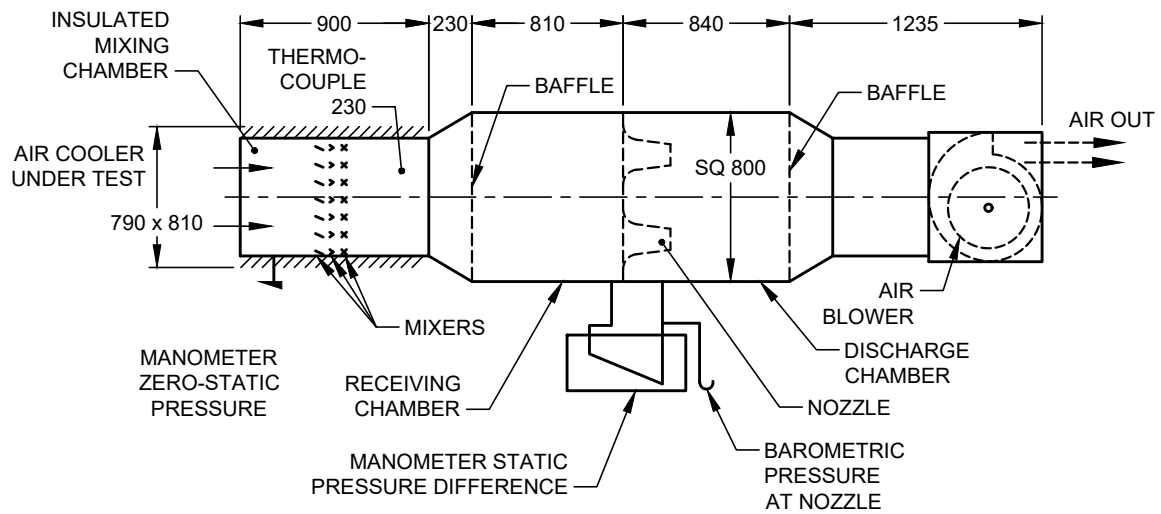


NOTE - LAB DIMENSIONS ARE AS PER THE MANUFACTURER'S REQUIREMENT WHERE THE TEST UNIT ROOM TEMPERATURES SHALL BE MAINTAINED.

FIG. 2 SUGGESTED SETUP FOR TESTING DIRECT EVAPORATIVE COOLERS

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(Page 8, Fig. 2) — Substitute the following for the existing:



All dimensions in millimetres.

FIG. 3 TYPICAL AIRFLOW AND TEMPERATURE MEASURING APPARATUS

(Page 8, Fig. 3) — Substitute the following for the existing:

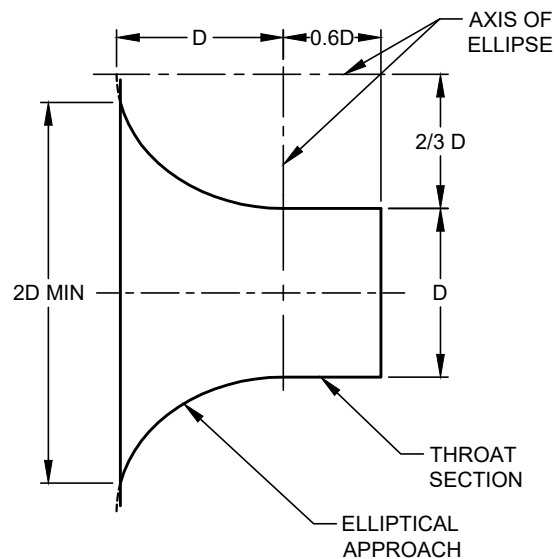


FIG. 4 AIRFLOW MEASUREMENT NOZZLE

(Page 8, clause 10.5) — Substitute 'Fig. 4' for 'Fig. 3'.

(Page 11, clause 14) — Substitute the following for the existing:

14 STORAGE

'It is recommended to store the air cooler in such a manner that there is no adverse effect on the functionality of air cooler.'

(Page 12, Annex A) — Insert the following at the end:

IS No.	Title
'IS/IEC 60529 : 2001	Degrees of protection provided by enclosures (IP CODE)
IS/IEC 60730-1 : 1999	Automatic electrical controls for household and similar use: Part 1 General requirements'.

(Page 14, Annex C) — Substitute 'Annex D' for 'Annex C'

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(Page 14, Annex C) — Insert the following new Annex C:

‘ANNEX C

(Clause 8.6.3)

EXAMPLE CALCULATION FOR TEST AND RATED CONDITIONS

C-1 CALCULATION ASSUMPTIONS

C-1.1 Calculation assumptions are listed below:

- a) Rated conditions (as per 8.6.3):
 - i) Rated inlet air DBT: 40 °C; and
 - ii) Rated inlet air WBT: 24 °C.
- b) Exemplary test conditions in the laboratory: (Test conditions as per 7);
 - i) Inlet air DBT: 36 °C; and
 - ii) Inlet air WBT: 24 °C.
- c) Airflow (measured dry): 1 500 m³/h;
- d) Measured outlet air DBT (during test conditions): 28 °C;
- e) Measured power input (during test conditions): 150 W;
- f) Air density: 1.127 kg/m³ at 40 °C DBT; and
- g) Specific heat of air (constant): 1.006 9 kJ/kg °C at 40 °C DBT.

C-2 TEST PROCEDURES AND CALCULATIONS AS PER IS 3315

C-2.1 Airflow Measurement (Dry Conditions) (See 8.3)

- a) Procedure — Set up the air cooler in the test chamber as per 10. Install airflow measurement apparatus (see 10.1). Ensure that the evaporating medium is dry (see 8.3.3). The pump is not operational. Run the cooler at maximum fan speed;
- b) Elaboration — This measurement establishes the baseline airflow the cooler provides without evaporative cooling. It is a prerequisite before starting cooling measurements, and the value is used later in the process. (see 8.3.1); and
- c) Measured Value: 1 500 m³/h.

C-2.2 Evaporation Efficiency Measurement and Calculations (See 7, 8.4, and 10)

- a) Procedure: Establish the 'test conditions' in the test chamber. The test room should conform to 10, which specifies requirements for instruments. In accordance with 8.6.2 and 8.6.3, the chamber shall be such that the average value is adopted given in 8.3.3 and 8.4.3. Additionally, the chamber meets requirements for a minimum wet-bulb depression of 11 °C for testing with a minimum dry-bulb temperature of 35 °C;
- b) Exemplary test Conditions are: 36 °C DBT and 24 °C WBT which meet the above requirements. Once stable operating conditions are achieved (see 7.8), measure the outlet air dry bulb temperature (T_{a2}) and power input. Ensure all conditions as specified in 8.4.3 are met. In this example, the measured outlet DBT is 28 °C. Power input measured at the test conditions is 150 W; and
- c) Calculate Evaporation Efficiency:

Calculate evaporation efficiency is given by below formula:

$$\eta = [T_{a1} - T_{a2} / T_{a1} - T_{w1}] \times 100$$

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where

$$\begin{aligned}T_{a1} &= 36\text{ }^{\circ}\text{C (Inlet DBT at Test conditions);}\\T_{a2} &= 28\text{ }^{\circ}\text{C (Measured outlet DBT at test conditions); and}\\T_{w1} &= 24\text{ }^{\circ}\text{C (Inlet WBT at Test conditions).}\end{aligned}$$

Evaporation efficiency calculation example is below:

$$\begin{aligned}\eta &= [(36 - 28)/(36 - 24)] \times 100\% \\ \eta &= [8/12] \times 100\% \\ \eta &= 66.67\%\end{aligned}$$

C-2.3 Cooling Capacity Calculation at Rated Conditions

Cooling capacity calculation at rated conditions are below:

- a) Rated Conditions: As per IS 3315, the air cooler ratings are declared for DBT of 40 °C and WBT of 24 °C. All measurements are then converted to calculations at rated conditions for the cooling capacity; and
- b) Calculation of Cooling Capacity:

Formula:

$$\text{Cooling capacity} = (\text{Airflow} \times \text{Density} \times \text{Air Heat Capacity} \times \text{Evaporation Efficiency} \times (\text{Rated inlet DBT condition} - \text{Rated Inlet WBT condition}) / 3\,600$$

where

$$\begin{aligned}\text{Measured airflow} &: 1\,500\text{ m}^3/\text{h} \\ \text{Density (at } 40\text{ }^{\circ}\text{C test)} &: 1.127\text{ kg/m}^3 \\ \text{Air heat capacity (at } 40\text{ }^{\circ}\text{C test)} &: 1.006\,9\text{ kJ/kg}\cdot^{\circ}\text{C} \\ \text{Evaporation efficiency} &: 66.67\% \text{ or } 0.666\,7 \\ (T_{a1} - T_{a2}) \text{ at rated conditions} &: 40\text{ }^{\circ}\text{C} - 24\text{ }^{\circ}\text{C} \text{ or } 16\text{ }^{\circ}\text{C}\end{aligned}$$

- c) Calculation Example:

$$\text{Cooling capacity} = (1\,500 \times 1.127 \times 1.006\,9 \times 0.666\,7 \times 16) / 3\,600 = 5.043\text{ kW}$$

C-2.4 Cooler EER Calculation at Rated Conditions

- a) Formula of cooler EER calculation at rated conditions is given below:

$$\text{EER} = \text{Cooling capacity (kW)} / \text{Power input (kW)}$$

where

$$\begin{aligned}\text{Sensible cooling capacity} &= 5.043\text{ kW} \\ \text{Power input} &= 0.15\text{ kW}\end{aligned}$$

- b) Calculation for EER is given below:

$$\text{EER} = 5.043/0.15 = 33.62'$$